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DevOps Shack

Corporate DevOps Workbook

Introduction

Welcome to the **Corporate DevOps Workbook**—your **go-to resource** for mastering daily DevOps operations. Whether you're a **DevOps engineer, SRE, or system administrator**, this guide provides a **comprehensive reference** for managing infrastructure, automating deployments, and troubleshooting issues.

This workbook covers critical **commands, workflows, and best practices** across industry-standard tools such as:

- ✓ **Git** – Version Control & Collaboration
- ✓ **Docker** – Containerization & Image Management
- ✓ **Kubernetes** – Container Orchestration & Scaling
- ✓ **Terraform** – Infrastructure as Code (IaC)
- ✓ **Azure DevOps** – CI/CD Pipelines & Automation
- ✓ **Linux** – System Administration & Networking

Why This Workbook?

- ◆ **Quick & Easy Access** – A single source for the most used DevOps commands.
- ◆ **Practical Use Cases** – Commands are structured with real-world applications.
- ◆ **Troubleshooting & Optimization** – Common issues and solutions for DevOps workflows.
- ◆ **Security & Best Practices** – Safe usage guidelines for each tool to avoid critical mistakes.

Below is a **quick reference table** featuring the **Top 40 Most Used DevOps Commands** categorized by tool, with **color-coded indicators** for command safety:

- ☐ **Safe Commands** – Everyday usage, minimal risk.
- ☐ **Caution Commands** – Requires attention, potential risk.
- **Destructive Commands** – High risk, irreversible actions.

Whether you're working on **deployments, infrastructure provisioning, or troubleshooting**, this workbook will help you **increase efficiency and reduce downtime**. Let's dive in! 🚀

Top 40 Most Used DevOps Commands (Quick Reference)

◆ Git (Version Control)

Command	Description	Safety
<code>git status</code>	Check the status of working directory	☐
<code>git pull origin <branch></code>	Fetch and merge latest changes from remote	☐
<code>git add .</code>	Stage all modified files for commit	☐
<code>git commit -m "message"</code>	Commit staged changes with a message	☐
<code>git push origin <branch></code>	Push local commits to remote repository	☐
<code>git checkout -b <branch></code>	Create and switch to a new branch	☐
<code>git merge <branch></code>	Merge specified branch into the current branch	☐
<code>git rebase <branch></code>	Reapply commits on top of another branch	☐
<code>git reset --soft <commit></code>	Undo commits but keep changes staged	☐
<code>git reset --hard <commit></code>	WARNING: Reset to a previous commit, losing all changes	●

Docker (Containers & Images)

Command	Description	Safety
<code>docker ps</code>	List running containers	☐
<code>docker ps -a</code>	List all containers (running & stopped)	☐
<code>docker images</code>	List all available Docker images	☐
<code>docker run -d -p 8080:80 <image></code>	Run a container in detached mode with port mapping	☐
<code>docker exec -it <container-id> bash</code>	Open shell inside a running container	☐
<code>docker logs <container-id></code>	View logs of a running container	☐
<code>docker stop <container-id></code>	Stop a running container	☐
<code>docker rm <container-id></code>	WARNING: Remove a stopped container	●
<code>docker rmi <image></code>	WARNING: Delete a Docker image	●
<code>docker system prune -a</code>	WARNING: Remove unused images, containers, and networks	●

Kubernetes (K8s)

Command	Description	Safety
<code>kubectl get pods</code>	List all running pods	
<code>kubectl describe pod <pod-name></code>	Get detailed information about a pod	
<code>kubectl logs <pod-name></code>	View logs of a pod	
<code>kubectl get deployments</code>	List all deployments	
<code>kubectl scale deployment <name> --replicas=3</code>	Scale deployment to 3 replicas	
<code>kubectl rollout status deployment <name></code>	Check deployment rollout status	
<code>kubectl exec -it <pod-name> -- /bin/sh</code>	Access a running pod's shell	
<code>kubectl delete pod <pod-name></code>	WARNING: Delete a specific pod	
<code>kubectl delete deployment <name></code>	WARNING: Remove a deployment	
<code>kubectl drain <node></code>	WARNING: Prepare a node for maintenance by evicting pods	

Terraform (Infrastructure as Code)

Command	Description	Safety
<code>terraform init</code>	Initialize Terraform working directory	☐
<code>terraform fmt</code>	Format Terraform configuration files	☐
<code>terraform validate</code>	Validate Terraform configuration files	☐
<code>terraform plan</code>	Preview changes before applying them	☐
<code>terraform apply</code>	Apply the Terraform configuration	☐
<code>terraform refresh</code>	Update Terraform state file with real infrastructure data	☐
<code>terraform destroy</code>	WARNING: Destroy all Terraform-managed resources	●
<code>terraform state list</code>	List all managed resources	☐
<code>terraform state show <resource></code>	Show details of a specific resource	☐
<code>terraform force-unlock <id></code>	WARNING: Manually unlock Terraform state (use with caution)	●

Linux & Shell Commands

Command	Description	Safety
<code>ls -la</code>	List files and directories with detailed information	☐
<code>cd <directory></code>	Change directory	☐
<code>mkdir <directory></code>	Create a new directory	☐
<code>rm -rf <directory></code>	WARNING: Remove a directory and its contents permanently	☒
<code>chmod +x <file></code>	Change file permissions to executable	☐
<code>chown user:group <file></code>	Change file ownership	☐
<code>ps aux</code>	List running processes	☐
<code>kill -9 <PID></code>	WARNING: Forcefully terminate a process	☒
<code>netstat -tulnp</code>	Show active network connections	☐
<code>tail -f /var/log/syslog</code>	View system logs in real-time	☐

Next Set of DevOps Commands

Introduction

Now that we've covered the Top 40 Most Used DevOps Commands, let's dive deeper into specific tools and workflows.

In the next sections, you'll find essential daily commands for:

- ✓ Git – Version Control
- ✓ Docker – Container Management
- ✓ Kubernetes – Orchestration
- ✓ Terraform – Infrastructure as Code
- ✓ Azure DevOps – CI/CD & Pipelines
- ✓ Linux – System Administration

Each section includes:

- ✂ Frequently Used Commands
- ✂ Real-World Use Cases
- ✂ Troubleshooting Tips

These commands will serve as a quick reference guide for DevOps engineers to efficiently manage deployments, infrastructure, and automation. Let's get started! 🚀 🔥

1. Git & Version Control

Basic Commands

Command	Description
git init	Initialize a new Git repository
git clone <repo-url>	Clone an existing repository
git status	Show status of working directory

Command	Description
git add <file>	Stage changes for commit
git commit -m "message"	Commit staged changes
git push origin <branch>	Push commits to a remote repository
git pull origin <branch>	Fetch and merge changes from remote
git log --oneline	Show commit history in short format
git diff	Show differences in modified files
git stash	Temporarily save changes without committing

Branching & Merging

Command	Description
git branch	List all branches
git checkout -b <branch>	Create and switch to a new branch
git merge <branch>	Merge specified branch into current branch
git rebase <branch>	Reapply commits on top of another branch
git branch -d <branch>	Delete a local branch

Reverting & Resetting

Command	Description
git reset --hard <commit>	Reset repository to a specific commit
git revert <commit>	Undo changes by creating a new commit
git checkout -- <file>	Discard changes in a working directory

2. Docker & Containerization

Basic Commands

Command	Description
<code>docker --version</code>	Show Docker version
<code>docker ps</code>	List running containers
<code>docker ps -a</code>	List all containers (running & stopped)
<code>docker images</code>	List all available images
<code>docker build -t <image-name> .</code>	Build a Docker image from Dockerfile
<code>docker run -d -p 8080:80 <image></code>	Run a container in detached mode with port mapping
<code>docker stop <container-id></code>	Stop a running container
<code>docker restart <container-id></code>	Restart a container
<code>docker logs <container-id></code>	View logs of a running container
<code>docker exec -it <container-id> bash</code>	Access a running container's shell

3. Kubernetes (K8s)

Pod Management

Command	Description
kubectl get pods	List all running pods
kubectl describe pod <pod-name>	Show details of a pod
kubectl logs <pod-name>	Fetch logs from a pod
kubectl delete pod <pod-name>	Delete a pod
kubectl exec -it <pod-name> -- /bin/sh	Access a running pod's shell

Deployments & Scaling

Command	Description
kubectl get deployments	List all deployments
kubectl create deployment <name> --image=<image>	Create a deployment
kubectl scale deployment <name> --replicas=3	Scale deployment to 3 replicas
kubectl rollout status deployment <name>	Check deployment rollout status
kubectl delete deployment <name>	Delete a deployment

4. Terraform (IaC - Infrastructure as Code)

Command	Description
terraform init	Initialize Terraform working directory
terraform fmt	Format Terraform files
terraform validate	Validate Terraform configuration
terraform plan	Show execution plan before applying
terraform apply	Apply the Terraform configuration
terraform destroy	Destroy all Terraform-managed infrastructure
terraform state list	List all managed resources
terraform state show <resource>	Show details of a specific resource
terraform output	Show Terraform outputs
terraform refresh	Sync state with real infrastructure

5. Azure DevOps & CI/CD Pipelines

Repositories

Command	Description
az repos list	List all repositories
az repos create --name <repo>	Create a new repository

git push --set-upstream origin <branch> Push a new branch to Azure Repos

Pipelines & Releases

Command	Description
az pipelines list	List all pipelines
az pipelines run --name <pipeline>	Run a specific pipeline
az artifacts list	List stored artifacts
az pipelines releases list	List release pipelines
az pipelines variable-group list	List all variable groups

6. Linux & Shell Scripting

File & Directory Management

Command	Description
ls -la	List files with details
cd <directory>	Change directory
mkdir <directory>	Create a new directory
rm -rf <directory>	Remove directory and its contents

User & Permission Management

Command	Description
whoami	Show current user
chmod +x <file>	Change file permissions
chown user:group <file>	Change file ownership

Process & Networking

Command	Description
ps aux	List running processes
kill -9 <PID>	Terminate a process
netstat -tulnp	Show active network connections

7. Monitoring & Logging

Prometheus & Grafana

Command	Description
<code>kubectl get pods -n monitoring</code>	List monitoring stack pods
<code>kubectl logs <pod-name> -n monitoring</code>	View Prometheus logs
<code>kubectl port-forward svc/grafana 3000:3000 -n monitoring</code>	Access Grafana

Log Management with ELK Stack

Command	Description
<code>curl -XGET "http://localhost:9200/_cat/indices?v"</code>	List Elasticsearch indices
<code>tail -f /var/log/syslog</code>	View system logs in real-time

8. Database & SQL Operations

Basic Commands

Command	Description
<code>mysql -u root -p</code>	Login to MySQL database
<code>SHOW DATABASES;</code>	List all databases
<code>USE <database>;</code>	Select a database
<code>SHOW TABLES;</code>	List all tables in the database
<code>SELECT * FROM <table>;</code>	Retrieve data from a table
<code>mysqldump -u user -p database > backup.sql</code>	Backup a MySQL database
<code>psql -U postgres -d mydb</code>	Connect to PostgreSQL
<code>SELECT COUNT(*) FROM <table>;</code>	Count records in a table
<code>DROP DATABASE <database>;</code>	Delete a database
<code>ALTER TABLE <table> ADD COLUMN <column> TYPE;</code>	Add a column to an existing table

Conclusion

The **Corporate DevOps Workbook** serves as a **comprehensive guide** for navigating daily DevOps operations efficiently. From **Git version control** to **container management with Docker and Kubernetes**, **infrastructure automation with Terraform**, and **CI/CD pipelines in Azure DevOps**, this resource equips engineers with **critical commands, troubleshooting techniques, and best practices** to streamline workflows.

Key Takeaways:

- ✓ **Efficiency Boost** – A single reference to execute DevOps tasks faster and with greater confidence.
- ✓ **Reduced Errors** – Color-coded safety indicators help prevent critical mistakes.
- ✓ **Troubleshooting Ready** – Includes solutions to common issues across multiple DevOps tools.
- ✓ **Security & Best Practices** – Guidelines to enhance security, automation, and operational resilience.

As DevOps continues to evolve, so should your skill set. Keep this workbook handy, update it with new findings, and use it as a **living document** to adapt to emerging technologies and best practices.